

Justin Lubin

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Curriculum Vitae, August 2026

Research Mission

To enable scientists to create the programs they need with *only* domain expertise, *not* programming expertise.

Research Areas: Programming languages ◦ human-computer interaction ◦ biology

Education

University of California, Berkeley 2020–

PhD in Computer Science (Major: Programming Languages, Minor: Bioinformatics)

Advisor: Sarah E. Chasins

Additional certification: Certificate in Teaching and Learning in Higher Education

University of Chicago 2016–2020

BS in Computer Science (Honors), BS in Mathematics, Minor in Music; *summa cum laude*

Advisor: Ravi Chugh

Honors Thesis: [Forging Smyth: The Impl. of Program Sketching with Live Bidirectional Evaluation](#)

Research Internships

Carnegie Mellon University 2018

Advisors: Jonathan Aldrich (CMU), Alex Potanin (Victoria University of Wellington)

Project: [Approximating Polymorphic Effects with Capabilities](#)

🏆 1st place, SPLASH Undergraduate Student Research Competition 2018

Peer-Reviewed Conference and Journal Publications

* = equal contribution, † = research mentee

Direct Manipulation and Natural Language Programming, Together at Last? OOPSLA '26

Parker Ziegler, David Minh-Duy Cao, **Justin Lubin**, and Sarah E. Chasins

To appear in *Proceedings of the ACM on Programming Languages (PACMPL)*, Issue OOPSLA2 (2026).

Transcriptional regulation of disease-relevant microglial activation programs Neuron '26

Amanda McQuade, Reet Mishra, Venus Hagan, Weiwei Liang, Peter J. Colias, Vincent Cele Castillo, Bianca Gonzalez, **Justin P. Lubin**, Verena Haage, Victoria Marshe, Masashi Fujita, Thomas Ta, Layla Gomes, Olivia Teter, Xia Han, Nathaniel Robichaud, Sarah E. Chasins, Jessica E. Rexach, Philip L. De Jager, James K. Nuñez, Martin Kampmann

In *Neuron* (2026).

Exploring the Translation-As-Explanation Folk Wisdom for Program Editing Tasks VL/HCC '26

Jacob Yim^{*†}, **Justin Lubin**^{*}, Kevin Ye[†], Laila Walker[†], Eric Rawn, and Sarah E. Chasins

To appear in *IEEE Symposium on Visual Languages and Human Centric Computing* (2026).

- Navigating AND–OR Graph Modifications to Debug Failing Proof Search** PLDI '26
Justin Lubin, Marlena Preigh[†], Max Willsey, and Sarah E. Chasins
 In *Proceedings of the ACM on Programming Languages (PACMPL)*, Issue PLDI (2026).
- Programming By Scaffolded Demonstration with *Perpend*** CHI '26
 Angela Bi*, Eric Rawn*, **Justin Lubin**, and Sarah E. Chasins
 In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (2026).
- Programmable epigenome editing by transient delivery of CRISPR epigenome editor ribonucleoproteins** Nat. Commun. '25
 Da Xu*, Swen Besselink*, Gokul N. Ramadoss, Philip H. Dierks, **Justin P. Lubin**, Rithu K. Pattali, Jinna I. Brim, Anna E. Christenson, Peter J. Colias, Izaiah J. Ornelas, Sarah E. Chasins, Bruce R. Conklin, and James K. Nuñez
 In *Nature Communications* (2025).
- Programming by Navigation** PLDI '25
Justin Lubin, Parker Ziegler, and Sarah E. Chasins
 In *Proceedings of the ACM on Programming Languages (PACMPL)*, Issue PLDI (2025).
 🏆 Selected as research highlight at [MIT Programming Languages Review 2026](#)
- Fast Direct Manipulation Programming with Patch-Reconciliation Correspondence** PLDI '25
 Parker Ziegler, **Justin Lubin**, and Sarah E. Chasins
 In *Proceedings of the ACM on Programming Languages (PACMPL)*, Issue PLDI (2025).
- Equivalence by Canonicalization for Synthesis-Backed Refactoring** PLDI '24
Justin Lubin, Jeremy Ferguson^{*†}, Kevin Ye^{*†}, Jacob Yim^{*†}, and Sarah E. Chasins
 In *Proceedings of the ACM on Programming Languages (PACMPL)*, Issue PLDI (2024).
- Exploring the Learnability of Program Synthesizers by Novice Programmers** UIST '22
 Dhanya Jayagopal^{*†}, **Justin Lubin**^{*}, and Sarah E. Chasins
 In *Proceedings of the ACM Symposium on User Interface Software and Technology* (2022).
- How Statically-Typed Functional Programmers Write Code** OOPSLA '21
Justin Lubin and Sarah E. Chasins
 In *Proceedings of the ACM on Programming Languages (PACMPL)*, Issue OOPSLA (2021).
 🏆 3rd place, CHI Graduate Student Research Competition 2021
- Program Sketching with Live Bidirectional Evaluation** ICFP '20
Justin Lubin, Nick Collins, Cyrus Omar, and Ravi Chugh
 In *Proceedings of the ACM on Programming Languages (PACMPL)*, Issue ICFP (2020).
- Sketch-n-Sketch: Output-Directed Programming for SVG** UIST '19
 Brian Hempel, **Justin Lubin**, and Ravi Chugh
 In *Proceedings of the ACM Symposium on User Interface Software and Technology* (2019).
 🏆 Best Demo Honorable Mention, UIST 2019
- Deuce: A Lightweight User Interface for Structured Editing** ICSE '18
 Brian Hempel, **Justin Lubin**, Grace Lu, and Ravi Chugh
 In *Proceedings of the International Conference on Software Engineering* (2018).

Preprints

** = equal contribution*

Programmable CRISPRtune dissects the transcriptional repressive activity of MeCP2

Jinna I. Brim*, Izaiah J. Ornelas*, Peter J. Colias, Nikita S. Divekar, Da Xu, **Justin P. Lubin**, Sophia I. Ferrel, Luis Galán Palma, Mitzi G. Hernández Zamora, Rithu K. Pattali, Jameel J. McDaniel, Sarah E. Chasins, and James K. Nuñez

LEMONmethyl-seq: Targeted long-read DNA methylation profiling reveals dynamics of CRISPR epigenome editing and endogenous DNA methylation patterns

Anna E. Christenson, Nikita S. Divekar, **Justin P. Lubin**, Luis G. Palma, Peter J. Colias, Rithu K. Pattali, Da Xu, Akane Hubbard, Katie Lin, Ngan Phan, Bernardo Moreno, Sarah E. Chasins, S. John Liu, James K. Nuñez

Workshop Papers

** = equal contribution, † = research mentee*

Interactive Design Space Exploration for Sparse Accelerators

YArch '26

Evan Williams[†], **Justin Lubin**, Rubens Lacouture, and Olivia Hsu
Presented by Evan at *Young Architect Workshop*.

Interactive Compiler Scheduling with Strong Guarantees

LATTE '26

Evan Williams[†], **Justin Lubin**, Rubens Lacouture, and Olivia Hsu
Presented by Evan at *Workshop on Languages, Tools, and Techniques for Accelerator Design*.

Searching for Incidental Specifications

PLATEAU '23

Jeremy Ferguson^{*†}, Kevin Ye^{*†}, Jacob Yim^{*†}, and **Justin Lubin**
In *Proceedings of the Workshop on Evaluation and Usability of Programming Languages and Tools*.

Type-Directed Program Transformations for the Working Functional Programmer

PLATEAU '19

Justin Lubin and Ravi Chugh
In *Proceedings of the Workshop on Evaluation and Usability of Programming Languages and Tools*.

Funding Awarded

\$600,000 — NSF 22-632: Cyberinfrastructure for Sustained Scientific Innovation (CSSI) 2026–2029
Co-wrote grant proposal with PI Sarah E. Chasins. Awarded for *SciInterop: Cyberinfrastructure to Design, Debug, and Reuse Interoperable, AI-Assisted Scientific Analysis Pipelines*.

Aim: To apply my research on **Programming by Navigation** to enable experimental scientists across all fields of science create, debug, and reuse scientific analyses.

\$159,000 — NSF Graduate Research Fellowship Program (GRFP)

2020–2025

Awarded for *Type-Directed Program Transformations for the Working Functional Programmer*.

Selected Honors

MIT Programming Languages Review selection of <i>Programming by Navigation</i>	2026
Demetri Angelakos Memorial Achievement Award (UC Berkeley)	2026
Evergreen Undergraduate Research Award (UC Berkeley)	2026
Undergraduate Research Mentoring Award (UC Berkeley)	2023
Outstanding Graduate Student Instructor Award (UC Berkeley)	2022
3 rd Place at CHI Graduate SRC, <i>How Statically-Typed Functional Programmers Author Code</i>	2021
NSF Graduate Research Fellowship (GRFP)	2020–2025
Enrico Fermi Scholar (University of Chicago)	2020
Harper Award (University of Chicago)	2020
Sigma Xi Honor Society	2020–
Best Demo Honorable Mention at UIST, <i>Sketch-n-Sketch: Output-Directed Prog. for SVG</i>	2019
Student Marshal (University of Chicago)	2019–2020
Phi Beta Kappa Honor Society	2019–
1 st Place at SPLASH Undergraduate SRC, <i>Approximating Polymorphic Effects with Capabilities</i>	2018
Dean's List (University of Chicago)	2016–2020

Selected Presentations, Workshops, and Lectures

Invited External Presentations

Pacific Programming Interfaces Confab (PPIC) — Programming by Navigation with Honeybee	2026
MIT Programming Languages Review — Programming by Navigation	2026
Stanford Software Research Lunch — Programming by Navigation	2025
UCSD PL/HCI Graduate Seminar (Philip Guo) — Q&A on <i>Exploring the Learnability of Program Synthesizers by Novice Programmers</i>	2025
UCSC Languages, Systems, and Data Seminar — Programming by Navigation	2024
UPenn PLClub — How Statically-Typed Functional Programmers Write Code	2022
UCSD PL Seminar — How Statically-Typed Functional Programmers Write Code	2021

Workshops for Biologists

Chan Zuckerberg Biohub — Intro to Genomics Analysis using Honeybee	2026
◦ Over 40 attendees from Stanford, UCSE, UC Berkeley, and the Chan Zuckerberg Biohub	
Nuñez Lab — From .FASTQs to .SVGs: How to Make a Bioinformatics Pipeline	2025
Nuñez Lab — Wrangling Flow Cytometry Data	2024
Neurona Therapeutics — Visualizing Data in Python with the Grammar of Graphics	2023
Neurona Therapeutics — Getting Started with Python for Wet Lab Data Analysis	2023
Nuñez Lab — Vizualizing Data in Python with the Grammar of Graphics	2023
Nuñez Lab — Analysis and Visualization of RNA-seq Data	2023

Guest Lectures (UC Berkeley)

Building User-Centered Programming Tools — Statistics for the Working PL+HCI Researcher	2026
Building User-Centered Programming Tools — User-Centered Design	2023
Programming Languages and Compilers — Compiler Optimizations	2021

Teaching

Teaching Positions

UC Berkeley CS 164: Programming Languages and Compilers (TA)	Fall 2022
UC Berkeley CS 164: Programming Languages and Compilers (Head TA)	Fall 2021
🏆 Received Outstanding Graduate Student Instructor Award	
UChicago CMSC 22300: Functional Programming (TA)	Spring 2020
UChicago CMSC 16100: Honors Introduction to Programming I (Grader)	Fall 2018

Certificate in Teaching and Learning in Higher Education (UC Berkeley) 2026

This certification required two semesters as a teaching assistant, an original course syllabus, mid- and end-of-semester evaluations from students with my plans for improvement, classroom observation of (and feedback on) my teaching from a faculty member, and the following workshops and courses:

Enhancing Student Engagement and Participation (Workshop)	2024
Assessing Teaching and Learning (Workshop)	2023
How Students Learn (Workshop)	2022
Syllabus and Course Design (Workshop)	2022
Graduate Student Instructor Ethics Course	2022
Teaching Techniques for Computer Science Course	2021
Graduate Student Instructor Conference on Teaching	2021

Mentorship

UG = undergrad

PhD Student Research Mentees

Marlena Preigh (advisor: Sarah E. Chasins @ UC Berkeley)	2025–
◦ Co-authorship at PLDI '26	
Evan Williams (advisor: Olivia Hsu @ CMU)	2025–
◦ Co-mentored with Rubens Lacouture @ Stanford	
◦ First authorship at LATTE '26 and YArch '26	

MS Thesis Research Mentees

Jacob Yim (UG, MS @ UC Berkeley → PhD @ UCSD)	2022–2025
◦ MS Thesis: Translations Alone Do Not Help Programmers Work With Unfamiliar Abstractions	
◦ Co-first authorship at VL/HCC '26	
◦ Co-authorship at PLDI '24	
◦ Co-first authorship at PLATEAU '23	
Jeremy Ferguson (UG, MS @ UC Berkeley → Software Engineer @ Voleon)	2022–2024
◦ MS Thesis: Eliciting Domain Expertise Reduces Examples Needed for Program Synthesis	
◦ Co-authorship at PLDI '24	
🏆 2 nd place at POPL Undergraduate SRC '23, Synthesizing Vectorized Code via Verified Lifting	
◦ Co-first authorship at PLATEAU '23	
Dhanya Jayagopal (UG, MS @ UC Berkeley → Software Engineer @ Udemy)	2020–2022
◦ MS Thesis: Study of Program Synthesizers & Novice Programmers	
◦ Co-first authorship at UIST '22	

Additional Research Mentees

Sam Poder (UG @ UC Berkeley)	2025–2026
Laila Walker (UG @ UC Berkeley → PhD @ UW)	2024–2025
◦ Co-mentored with MS Thesis Research Mentee Jacob Yim	
◦ Co-authorship at VL/HCC '26	
Kevin Ye (UG, Gap-year Researcher @ UC Berkeley → MS @ Simon Fraser University)	2022–2024
◦ Co-authorship at VL/HCC '26	
◦ Co-authorship at PLDI '24	
◦ Co-first authorship at PLATEAU '23	
Fayaz Shaik (UG @ Ohlone College & TTE REU Mentee → UG @ UCSD)	2021–2022
◦ Poster at SACNAS NDiSTEM '21, <i>User-Centered Data Population of Knowledge Graphs</i>	

SIGPLAN-M Mentees

Carolina Carreira (PhD @ CMU Portugal)	2022–
Adharsh Kamath (Research Fellow @ MSR India, PhD @ UIUC)	2022–

Additional Mentees

Kasozi Vincent (MS @ University of Chlef)	2022–2023
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Professional Activities and Research Service

Dagstuhl Participation

Theories of Programming	2022
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Organizing Committee

Workshop on Evaluation and Usability of Programming Languages and Tools	PLATEAU '26
Workshop on Evaluation and Usability of Programming Languages and Tools	PLATEAU '25
Workshop on Evaluation and Usability of Programming Languages and Tools	PLATEAU '24
◦ Additional Responsibilities: Publicity Chair	
Workshop on Evaluation and Usability of Programming Languages and Tools	PLATEAU '23

Program Committee

Workshop on Human Aspects of Types and Reasoning Assistants	HATRA '26
Workshop on Live Programming	LIVE '26
Workshop on Human Aspects of Types and Reasoning Assistants	HATRA '25
Workshop on Live Programming	LIVE '25
Workshop on Human Aspects of Types and Reasoning Assistants	HATRA '24
Workshop on Live Programming	LIVE '24
Symposium on New Ideas in Programming and Reflections on Software (Papers Track)	Onward! '24
Workshop on Human Aspects of Types and Reasoning Assistants	HATRA '23
Workshop on Live Programming	LIVE '23

Reviewer

ACM Symposium on User Interface Software and Technology	UIST '25
🏆 Received Special Recognition for Outstanding Review (×2)	
ACM Conference on Human Factors in Computing Systems	CHI '25
Journal of Functional Programming	JFP '24

ACM Conference on Human Factors in Computing Systems	CHI '23
🏆 Received Special Recognition for Outstanding Review	
Formal Methods in System Design	FMSD '22
ACM Symposium on User Interface Software and Technology	UIST '21

Student Volunteer

PL+HCI “Swimmer” School (Social Sessions Co-organizer)	2022
Elm in the Spring Conference	2019
International Conference on Functional Programming (ICFP)	2018

Leadership and Community Service

Incarceration to College Program — Tutor	2025–
Tutoring incarcerated youth in mathematics to prepare for post-secondary education.	

Berkeley Underground Scholars — Graduate Tutor	2023–
Tutoring formerly-incarcerated/systems-impacted students in computer science and mathematics.	

EECS Peers — Co-organizer	2023–
Co-leading a graduate student peer support and mentorship organization. (<i>Mentor since 2022.</i>)	

Equal Access to Application Assistance Program — Reviewer	2023–
Reviewed PhD applications as part of a student-run initiative aiming to ensure that all PhD applicants to UC Berkeley have access to guidance on the application process.	

SIGPLAN-M Long-Term Mentorship Program — Mentor	2022–
Serving as a mentor for the SIGPLAN-M long-term programming languages mentorship program.	

Association of Women in EE&CS (AWE) — Grad School Advisor	2023
Hosted series of office hours to advise women and nonbinary undergraduates interested in grad school.	

WiCSE Girl Scout Engineering Day — Activity Designer and Leader	2023
Designed and led educational activities for the Girl Scouts of Northern California to earn their programming and robotics badge.	

Women in Computer Science and Electrical Engineering (WiCSE) — Grad Mentor	2022
Served as a mentor for the WiCSE Grad Mentorship Program, where elder PhD students get paired up with first- and second-year PhD students for general mentorship.	

#ShutdownPL — Co-organizer	2020–2022
Cooordinated logistics and managed funding for three keynote speakers at dedicated anti-racist events at ICFP and POPL.	

Transfer-to-Excellence Summer Research Program (TTE REU) — Mentor	2021
Served as a research mentor for the TTE REU, a research experience for community college students seeking to transfer to four-year universities.	

compileHer Tech Capstone — Activity Leader	2019
Led computer science educational activities for middle school girls in Chicago.	

Internal Service

UC Berkeley **Programming Systems (PS) Research Group** and **EECS Department**

EECS Faculty Hiring Committee, Student Member	2025, 2026
EECS Visit Days Peer Advisor	'21, '22, '23, '25, '26
PS Visit Days Coordinator	2022
PS Seminar Organizer	2021–2022
PS Graduate Admissions Representative	2021